Parul® University
Vadodara, Gujarat
NAAC GRADE A++

Information and
Communication Technology

UNIT 3 : Concept Learning

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Content

- **Formulation of Hypothesis**
- **Probabilistic Approximately Correct Learning**
- **VC Dimension**
- **Hypothesis elimination**
- **Candidate, Elimination Algorithm**

What is Concept Learning?

- Concept Learning is a Machine Learning approach where a system learns a concept from examples.
- The goal is to determine whether a new example belongs to a target class.

Example

A model learns whether a day is suitable for playing tennis based on:

- Weather
- Temperature
- Humidity
- Wind

Concept Learning

Key Objectives

- Learn from training examples
- Generalize unseen examples
- Build hypotheses for classification

Important Terms of Concept Learning:

Term	Meaning
Instance	Input example
Attribute	Property of data
Hypothesis	Possible rule describing data
Target Concept	Actual concept to be learned

Advantages of Concept Learning

1. Learns from Examples

Machine Learning Concept learning enables machines to learn patterns directly from training data.

2. Improves Prediction Accuracy

By learning concepts from examples, models can make accurate predictions on unseen data.

3. Reduces Human Effort

Automatic learning minimizes the need for manually creating rules.

4. Supports Generalization

The system can generalize knowledge from known examples to unknown situations.

Advantages of Concept Learning

5. Foundation of Supervised Learning

Concept learning forms the basis of many supervised machine learning algorithms.

6. Useful in Classification Problems

It is highly effective for:

- Spam detection
- Disease prediction
- Sentiment analysis
- Pattern recognition

Limitations of Concept Learning

1. Sensitive to Noisy Data

Incorrect or inconsistent training examples may lead to wrong hypotheses.

2. Large Hypothesis Space

As the number of attributes increases, the hypothesis space grows rapidly.

3. Computational Complexity

Searching and updating hypotheses may become computationally expensive.

4. Requires Labeled Data

Concept learning depends on supervised data with correct labels.

5. Overfitting Problem

The model may memorize training examples instead of learning generalized concepts.

6. Difficulty with Continuous Data

Traditional concept learning methods work better with discrete attributes than continuous values.

Applications of Concept Learning

1. Email Spam Filtering
Classifies emails as spam or non-spam.
2. Medical Diagnosis
Predicts diseases based on symptoms and medical reports.
3. Weather Prediction
Learns weather patterns from historical data.
4. Recommendation Systems
Used in:
- Netflix
- Amazon
- YouTube

to recommend products or content.

Application of Concept Learning

5. Image Classification
Recognizes objects, faces, or handwritten digits in images.

6. Fraud Detection
Identifies suspicious banking or financial transactions.

7. Speech Recognition
Converts spoken language into text using learned concepts.

8. Student Performance Prediction
Predicts academic outcomes using learning behavior data.

What is a Hypothesis?

A **hypothesis** is a clear, testable statement or prediction about the relationship between two or more variables.

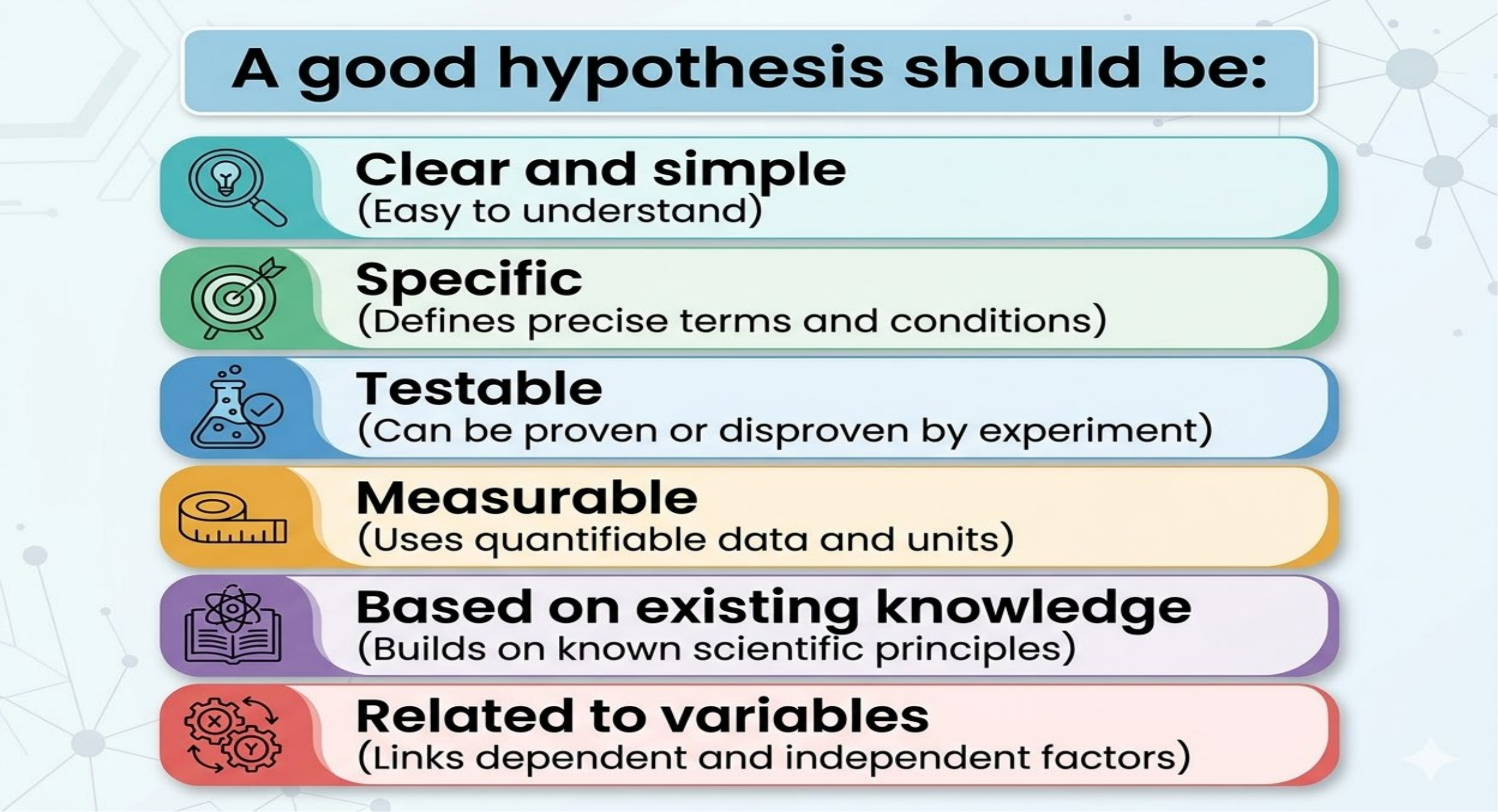
❑ **Key Points**

- Acts as a foundation for research
- Predicts possible outcomes
- It can be tested using experiments or data

❑ **Example**

"Students who study regularly score higher marks than students who study occasionally."

Characteristics of a Good Hypothesis



Types of Hypothesis

Type	Description	Example
Null Hypothesis (H_0)	No relationship between variables	“Study hours do not affect marks.”
Alternative Hypothesis (H_1)	Relationship exists	“Study hours improve marks.”
Simple Hypothesis	One independent and one dependent variable	“Exercise reduces stress.”
Complex Hypothesis	Multiple variables involved	“Diet and exercise improve health and productivity.”

Steps in Formulating a Hypothesis

1. Identify the research problem
2. Review related literature
3. Identify variables
4. Develop logical relationship
5. Write a clear and testable statement

Example of Hypothesis Formulation

Topic:
Effect of Social Media on Student Performance

Variables
•Independent Variable: Social media usage
•Dependent Variable: Academic performance

Hypothesis
"Excessive social media usage negatively affects students' academic performance."

PAC - Learning

- **PAC** stand for **Probably approximately correct**
- **Probably approximately correct (PAC) learning** is a framework for mathematical analysis of machine learning algorithm.
- In the other words PAC learning is a theoretical framework for analyzing the generalization performance of machine learning algorithms.

Goal: With High Probability ("Probably"),the select hypothesis will have lower error ("Approximately Correct"")

In the PAC model ,we specify two small parameter ,ε (epsilon) and δ (delta) and require that with probability at least (1-δ) a system learn a concept with error at most ε.

PAC - Learning

ε and δ parameters:

- ε gives an upper bound on the error in the accuracy with which h approximated(Accuracy : 1-ε)
- δ gives the probability of failure in the achieving this accuracy (Confidence : 1-δ)

PAC - Learning

- A good learner will learn with high probability and close approximation to the target concept

- With high probability, the selected hypothesis will have lower the error ("Approximately Correct") with the parameter ε and δ

PAC - Learning

PAC learning requires:

- small parameters ε (epsilon) and δ (delta),
- with probability at least (1 − δ), a system learns the concept with error at most ε.

➢ ε is the upper bound on the error in accuracy, i.e., the hypothesis with error less than ε.

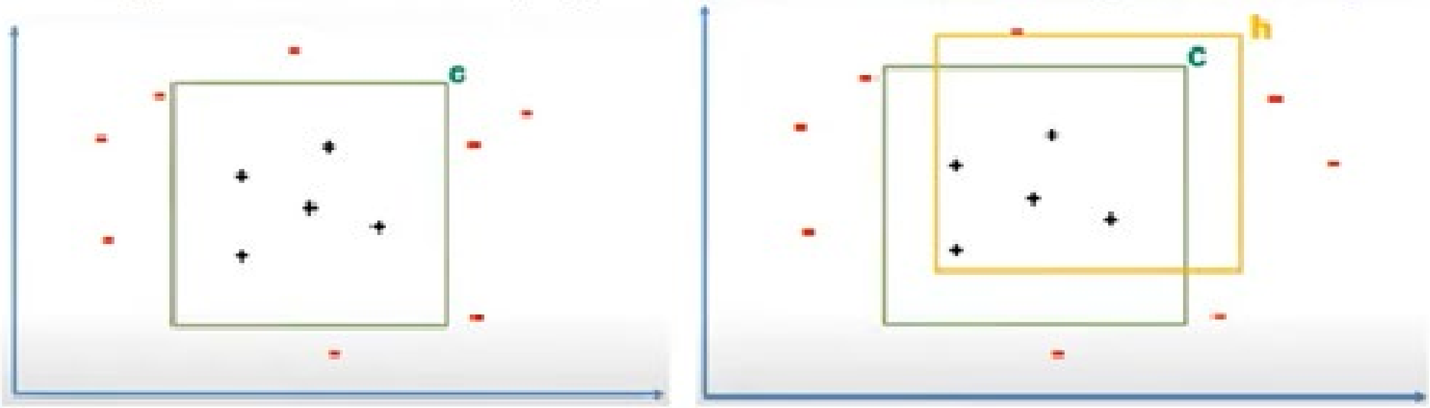
Accuracy = 1 − ε

➢ δ gives the probability of failure in achieving this accuracy.
For δ, where ($0 < \delta \le 1$), the hypothesis generated is approximately correct at least (1 − δ) of the time.

Confidence = 1 − δ

PAC – Learning Example

- N number of Car having Price and Engine power, as training set, (p,e), find the car is family car or not.
- An algorithm gives answer whether the car is family car or not.
- C – Target function
- Instances within rectangle represents family cars and outside are not family cars
- Hypothesis h – closely approximate C, and there may be error region.



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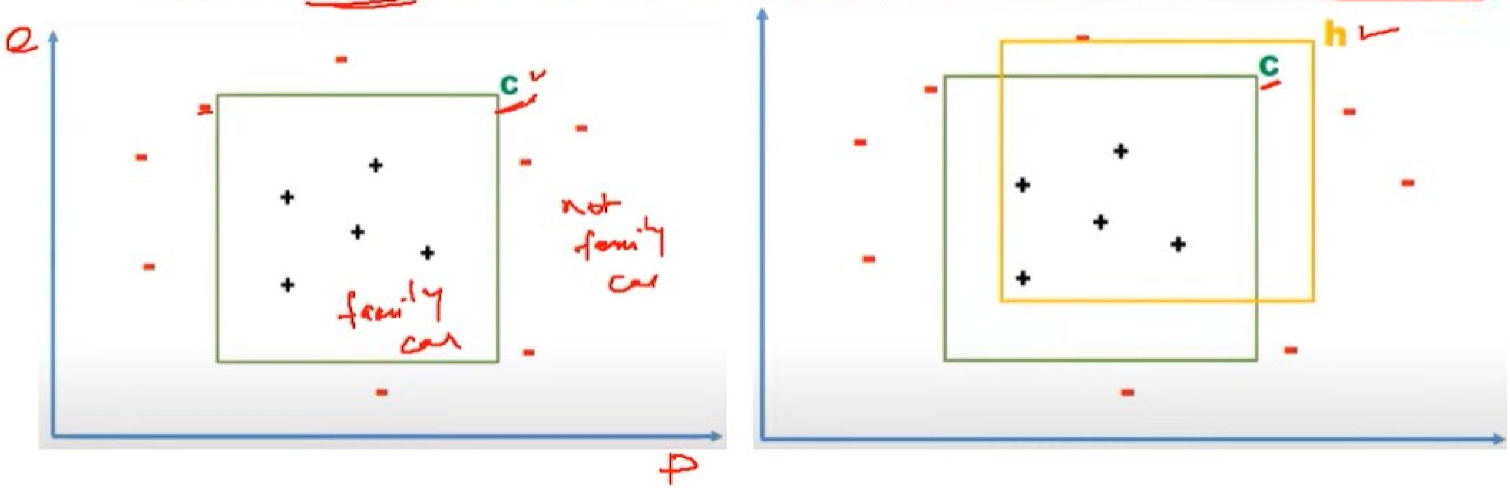
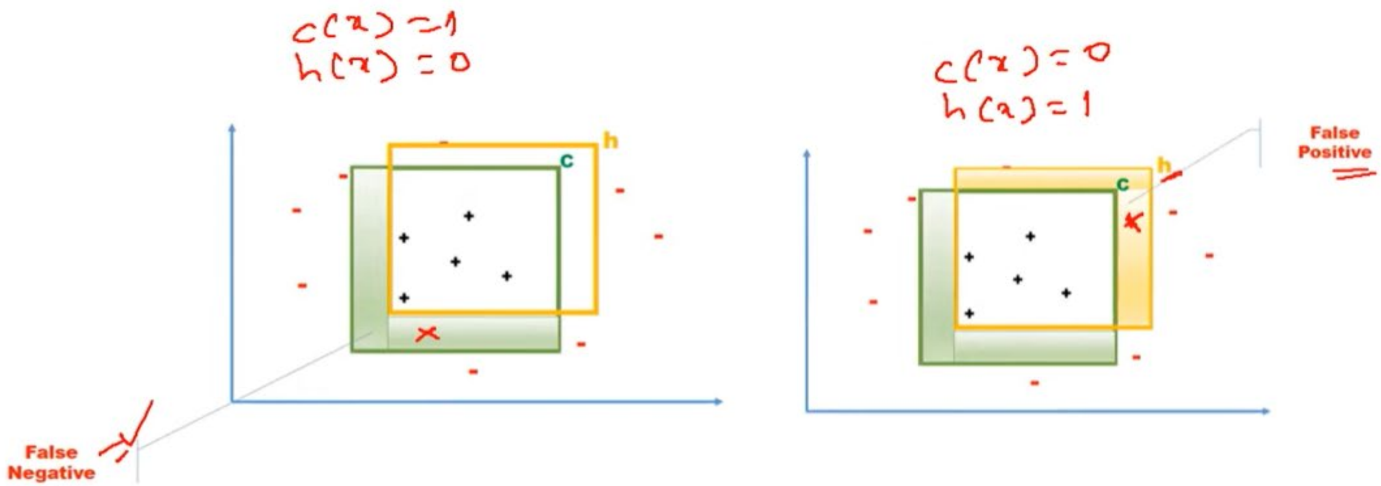


PAC – Learning Example

False Negative and False Positive

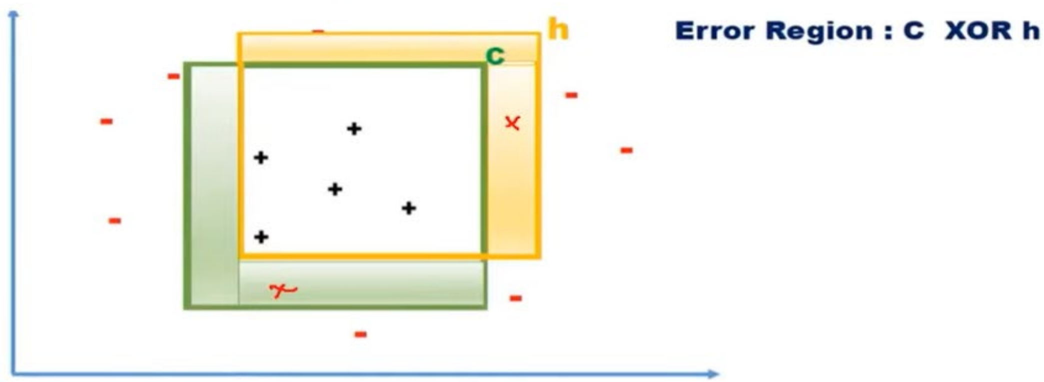
- Instances lies on shaded region are positive/negative according to our actual function ‘C’, but those are negative/positive based on the hypothesis h. Hence it is called as false negative or false positive



PAC – Learning Example

Error Region

- The probability of error region to be small
- The error region : $\boxed{P(C \text{ XOR } h) \le \varepsilon.}$ $p(c \oplus h)$



Probably Approximately Correct

Approximately Correct

•The hypothesis $h$is approximately correct, and the error is less than or equal to ε(epsilon).

•Where:

$$0 \leq \varepsilon \leq \frac{1}{2}$$

•i.e.,

$$\boldsymbol{P(C \oplus h) \leq \varepsilon}$$

•where:

- C= target concept
- h= learned hypothesis
- $\bigoplus$= XOR operation representing disagreement/error between $C$and h

Probably Approximately Correct

- Low generalization error with high probability.
- The probability that the error of hypothesis h is less than or equal to ε is at least $1-\delta$.

$$\boldsymbol{P(Error(h) \leq \varepsilon) \geq 1-\delta}$$

Equivalently,

$$\boldsymbol{P(P(C \oplus h) \leq \varepsilon) \geq 1-\delta}$$

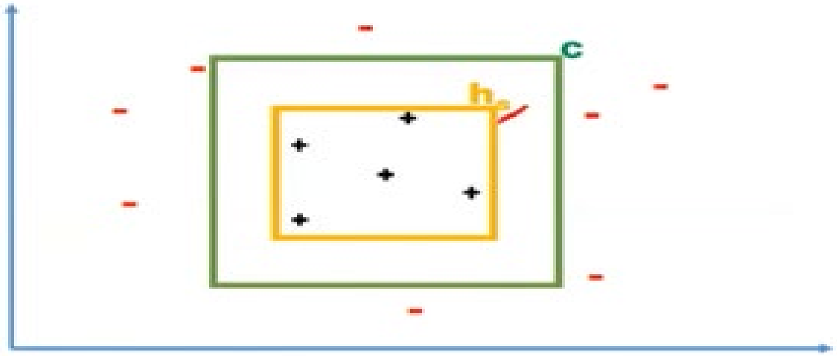
where:

- C = target concept
- h = learned hypothesis
- ε = maximum allowed error
- δ = probability of failure
- $1-\delta$ = confidence level

Probably Approximately Correct

PAC learnability for axis-aligned rectangle

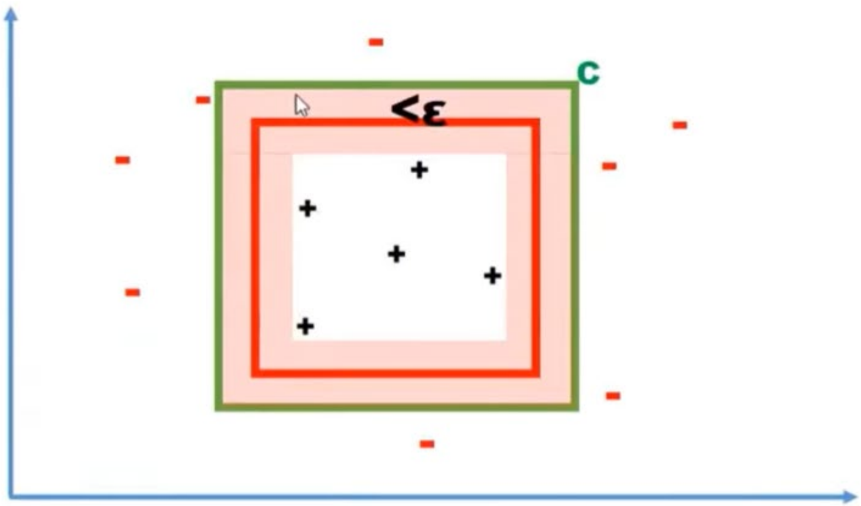
- Specialization:
- h_s is the tightest possible rectangle around a set of positive training examples.
- h_s is subset of C , Hence Error region = C - h



Probably Approximately Correct

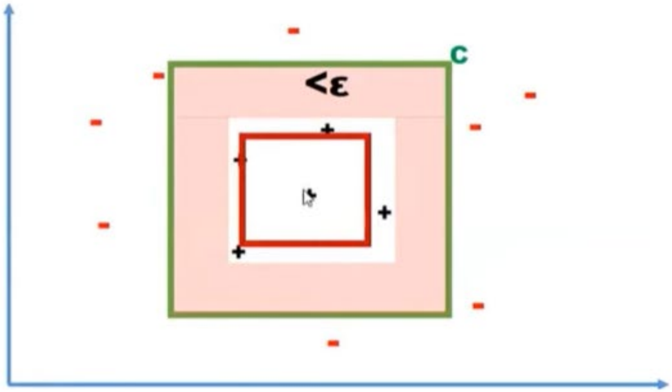
Approximately correct

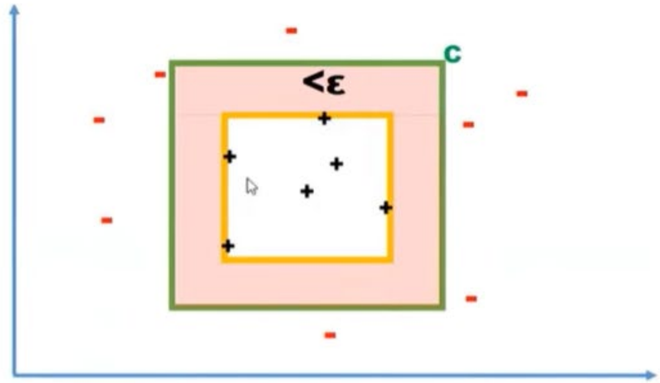
- If an hypothesis lies between h and c (shaded region) then it is approximately correct.



Probably Approximately Correct

- If the generated hypothesis does not touch any of these region
- Error region is greater than Ɛ and not approximately correct, because the error region got increased.
- Atleast one +ve example at each side of the rectangle

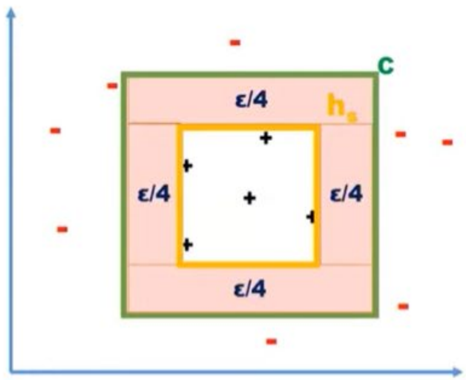




Probably Approximately Correct

Error Region

- Error Region = sum of four rectangular strips < Ɛ
- Each strip is at most Ɛ/4
- Probability of positive example falling in any one of the strip (error region = Ɛ/4)
- Probability that a randomly drawn positive example misses a strip = 1- Ɛ/4
- P(m instance miss a strip) = $(1-\mathcal{E}/4)^m$
- P(m instances miss any strip) < $4(1-\mathcal{E}/4)^m$
- Finally we get m > 4/ Ɛ log 4/ δ



PAC - Example

Example Problem 1

Sl.No.	Error(h1)
1	0.001
2	0.025
3	0.07
4	0.003
5	0.035
6	0.045
7	0.027
8	0.065
9	0.012
10	0.036

- Hypothesis h1 generated the errors with respect to price and engine power of given 10 samples,
- *Given*, $\varepsilon = 0.05$ $\delta = 0.20$
- $P(h1) >= 1-\delta$
- $P(h1)= 8/10 = 0.80$ (3rd and 8th values are greater than ε)
- Therefore, $0.80 >= (1 - 0.20)$ i.e. $0.80 = 0.80$
- **Hence h1 is probably approximately correct**

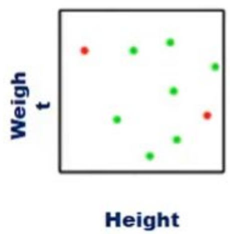
PAC - Example

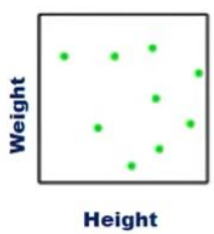
Example Problem 2

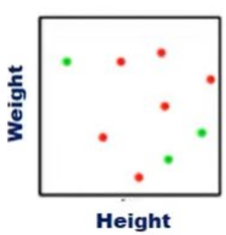
Sl.No.	Error(h2)
1	0.012
2	0.015
3	0.071
4	0.063
5	0.022
6	0.045
7	0.011
8	0.029
9	0.066
10	0.031

- Hypothesis h2 generated the errors with respect to price and engine power of given 10 samples,
- *Given*, $\varepsilon = 0.05$ $\delta = 0.20$
- $P(h2) >= 1-\delta$
- $P(h2)= 7/10 = 0.70$ ($3^{rd}, 4^{th}, 9^{th}$ values $> \varepsilon$)
- Here, $0.70 >= (1-0.20)$ i.e. $0.70 < 0.80$
- Hence h2 is not probably approximately correct

PAC - Example







Error(H1)	Error(h2)
0.04	0.04
0.03	0.035
0.09	0.039
0.06	0.06
0.025	0.025
0.049	0.059
0.04	0.04
0.03	0.03
0.05	0.55
0.043	0.043

$\varepsilon = 0.05$ $\delta = 0.20$

P (H1) = 8/10=0.80

P (H1) = 8/10=0.80 >=1- 0.20

Hence H1 is probably approximately correct

P(H2)= 7/10=0.70

P(H2)= 7/10=0.70 <1-0.20

Hence H2 is not probably approximately correct

What is VC Dimension?

VC Dimension is a measure of the **capacity or complexity** of a machine learning model.

Simple Definition,
“VC Dimension is the maximum number of points that can be perfectly shattered by a classifier.”

It tells us:
•How powerful a model is
•How many data points a model can classify correctly in all possible ways

Concept of Shattering

A set of points is said to be **shattered** if a classifier can separate them in **all possible label combinations**.

Example

For 3 points:

•Total label combinations = $2^3 = 8$

•If all 8 combinations can be classified correctly → points are shattered

Important Idea

Higher VC Dimension = More complex model

VC Dimension of Different Models

Model	VC Dimension
Point Classifier	1
Line in 2D	3
Circle in 2D	3
Neural Networks	Very High
Polynomial Models	Depends on degree

Explanation
•Simple models → Low VC Dimension
•Complex models → High VC Dimension

Importance of VC Dimension

- Helps measure model complexity
- Explains overfitting and underfitting
- Assists in model selection
- Improves generalization ability

Key Point:

- Very high VC Dimension → Overfitting risk
- Very low VC Dimension → Underfitting risk

Mathematical Representation

The VC Dimension is related to:

$$\boldsymbol{VC(H) = d}$$

Where:
•H= Hypothesis Space
•d= Largest number of points shattered

Example
A linear classifier in 2D can shatter:$3\ points$
But cannot shatter:$4\ points$

Concept Learning

Concept learning refers to the process of inferring a general description of a target concept from a set of examples. It is a fundamental component of many machine learning algorithms, especially in the context of supervised learning. Concept learning helps machines identify and classify objects, phenomena, or situations based on their characteristics.

The Concept Learning Task

The primary goal of concept learning is to find a hypothesis $h \in H$
such that,

$$\forall x,\ h(x) = C(x)$$

where x represents an instance, and $C(x)$ is the true classification.
In practice, the goal is to find h that closely approximates C
when C is unknown.

What is Hypothesis Elimination?

- Hypothesis Elimination is a concept in Machine Learning used to remove inconsistent hypotheses from the hypothesis space based on training examples.
- It helps in identifying the set of hypotheses that correctly classify all observed training examples.

- **Key Idea**
  - Start with all possible hypotheses
  - Eliminate hypotheses inconsistent with data
  - Retain only valid hypotheses

- **Applications**
  - Concept Learning
  - Classification Problems
  - Machine Learning Training Models

Hypothesis Space and Version Space

- **Hypothesis Space (H)**

  The complete set of all possible hypotheses that can describe the training data.

- **Version Space (VS)**

  The subset of hypotheses that are consistent with all training examples.

- **Important Components:**

1. Specific Boundary (S)

•Most specific hypothesis

•Represents minimum generalization

2. General Boundary (G)

•Most general hypothesis

•Represents maximum generalization

Version Space

A **Version Space** is the subset of all possible hypotheses in the hypothesis space H that are consistent with the training examples. A hypothesis is considered consistent if it correctly classifies all the training examples.

$$\textbf{Version Space (VS)} = \{h \in H \mid \forall (x,y) \in D,\ h(x)=y\}$$

where:
H: Hypothesis space
D: Training data
(x, y): An example x with label y

General and Specific Boundaries

The version space is defined by its **general boundary** (G) and **specific boundary** (S):

1. **General Boundary (G)**:
   The set of the most general hypotheses in H that are consistent with the training data. These hypotheses cover as many positive examples as possible without misclassifying negative examples.

2. **Specific Boundary (S)**:
   The set of the most specific hypotheses in H that are consistent with the training data. These hypotheses cover only the positive examples and nothing more.

The version space lies between S and G, encompassing all hypotheses that are more general than S and more specific than G.

Candidate Elimination Algorithm

The **Candidate Elimination Algorithm** refines S and G iteratively as new examples are presented. The goal is to narrow the version space until it converges to the target concept.

Steps of the Algorithm:

1) **Initialization**:
   Start with S as the most specific hypothesis in H.
   Start with G as the most general hypothesis in H.

2) **For each training example (x, y):**
   If y is positive:
      Remove from G any hypothesis that fails to classify x as positive.
      Generalize S as minimally as possible to include x, while ensuring consistency with all prior examples.

Candidate Elimination Algorithm

- If y is negative:

  Remove from SSS any hypothesis that incorrectly classifies x as positive.
  Specialize G as minimally as possible to exclude x, while ensuring consistency with all prior examples.

3) **Stop**:

   When S and G converge to a single hypothesis, that hypothesis is the learned target concept.
   If S and G become disjoint, the data is inconsistent.

Example

Consider a concept learning task with the following hypothesis space:

Feature	Values
Color	Red, Green, Blue
Shape	Circle, Triangle, Square
Size	Small, Medium, Large

Training Examples:

Example	Color	Shape	Size	Class
1	Red	Circle	Small	Positive
2	Green	Circle	Medium	Negative
3	Red	Triangle	Small	Positive

Example

Step-by-step process:

1. **Initialization**:
   S=[Red, Circle, Small] (most specific).
   G=[?, ?, ?] (most general).

2. **Processing Example 2** (Negative):
   S remains unchanged as it doesn't classify the negative example.
   G is specialized to exclude [Green, Circle, Medium], yielding:
      [Color ≠ Green, ?, ?]
      [?, Shape ≠ Circle, ?]
      [?, ?, Size ≠ Medium].

3. **Processing Example 3** (Positive):
   S is generalized to [Red, ?, Small to include the new positive example.
   G is adjusted for consistency, refining further if necessary.

This process continues until S and G converge.

Advantages of Candidate Elimination

- Systematically narrows the hypothesis space.
- Guarantees the target concept lies within S and G if data is noise-free.

Limitations

- Computationally expensive for large hypothesis spaces.
- Sensitive to noisy or inconsistent data.
- Requires a finite hypothesis space.

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